CLASS 10 MATHEMATICS • CHAPTER 6

Triangles

Exhaustive point-by-point presentation covering similarity of figures, Thales theorem, similarity criteria, and applications directly from the textbook.

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Introduction & Indirect Measurements

Congruence vs. Similarity

- **Congruent figures:** Same shape and same size.
- **Similar figures:** Same shape but not necessarily the same size.
- Focus of this chapter: Similarity of triangles and its applications (e.g., Pythagoras Theorem).

Indirect Measurements

- How do we measure mountains like Everest or distances to the moon?
- Direct tape measurement is impossible!
- We rely on **indirect measurements** based on the principle of similarity of figures.

Section 6.1 Live Check

Q1. What is the fundamental difference between congruent figures and similar figures?

Q2. Can you name a technique used to measure massive objects or distant bodies like Mount Everest or the moon?

Q3. Are all congruent figures also similar? Are all similar figures congruent?

Similar Figures & Conditions for Polygons

Examples of Similar Figures

- All circles are similar (radii may differ).
- All squares are similar.
- All equilateral triangles are similar.
- Photographs of the same monument in different sizes are similar.

Conditions for Polygon Similarity

Two polygons with the same number of sides are similar if:

- 1. All corresponding angles are equal.
- 2. All corresponding sides are in the same ratio (proportion / scale factor).

Section 6.2 Live Check

Q1. State the two essential conditions required for two polygons of the same number of sides to be similar.

Q2. Why are a square and a rhombus with sides of 3 cm not necessarily similar? Which condition fails?

Q3. What do we call the ratio of corresponding sides of two similar polygons?

Basic Proportionality Theorem (Thales Theorem)

Theorem 6.1 (Basic Proportionality Theorem)

If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio.

In $\triangle ABC$, if $DE \parallel BC$, then: $\frac{AD}{DB} = \frac{AE}{EC}$

Theorem 6.1 Live Check

Q1. Who was Thales, and what fundamental geometric theorem is attributed to him?

Q2. State the Basic Proportionality Theorem in your own words.

Q3. When a line is parallel to one side of a triangle cutting the other two sides, what equal area ratio is used in the proof?

Theorem 6.2 (Converse of Basic Proportionality Theorem)

Theorem 6.2 Statement

If a line divides any two sides of a triangle in the same ratio, then the line is parallel to the third side.

Proof Technique

Proved using **proof by contradiction**. Assume line DE is not parallel to BC, draw another parallel line DE', and show that E and E' must coincide.

Application

Used to test whether lines connecting points on sides of triangles are parallel by checking segment ratio equality.

Theorem 6.2 Live Check

Q1. What is the converse of the Basic Proportionality Theorem? State it clearly.

Q2. What proof method is employed to establish Theorem 6.2?

Q3. If a line divides sides AB and AC such that AD/DB = AE/EC, what conclusion can we draw about line DE?

AAA & AA Similarity Criteria

Theorem 6.3 (AAA Criterion)

- If in two triangles, corresponding angles are equal, then their corresponding sides are in the same ratio (proportional).
- Therefore, the two triangles are **similar**.

AA Criterion

- If two angles of one triangle are respectively equal to two angles of another triangle, the third angles are automatically equal.
- Thus, **AA similarity** is sufficient.

AAA & AA Criteria Live Check

Q1. What does AAA stand for in the context of triangle similarity?

Q2. Why is knowing just two pairs of corresponding angles equal enough to ensure similarity (AA criterion)?

Q3. If two triangles are equiangular, what can we say about the ratio of their corresponding sides?

SSS & SAS Similarity Criteria

Theorem 6.4 (SSS Criterion)

- If the sides of one triangle are proportional to the sides of another triangle, their corresponding angles are equal.
- The triangles are similar.

Theorem 6.5 (SAS Criterion)

- If one angle of a triangle equals one angle of another, and the sides including these angles are proportional, they are similar.

SSS & SAS Criteria Live Check

Q1. State the SSS similarity criterion in your own words.

Q2. What does SAS stand for, and what specific position must the proportional sides occupy relative to the equal angle?

Q3. Do we need to check both angle equality and side proportionality when using SSS or SAS? Why or why not?

Worked Exercise Highlights

Problem 1: Trapezium Ratios (Ex 6.2 Q9)

- Trapezium ABCD with AB || DC, diagonals intersect at O.
- Draw line through O parallel to AB.
- Apply Thales theorem in triangles DAB and CAB.
- **Proved: AO / BO = CO / DO.**

Problem 2: Lamp-post Shadow (Ex 6.3 Q7)

- Girl of height 0.9m walks at 1.2 m/s away from 3.6m lamp-post for 4s.
- Shadow length after 4s = x meters. Distance walked = 4.8m.
- By AA similarity of small and large triangles: (4.8 + x) / x = 3.6 / 0.9.
- **Shadow length x = 1.6 m.**

Exercise Highlights Live Check

Q1. In trapezium diagonal ratio proofs, why do we construct a line parallel to the bases through the intersection point?

Q2. How do we compute the distance walked by the girl in shadow problems given her speed and time?

Q3. Which similarity criterion is most frequently used when dealing with shadow and lamp-post problems?

Chapter Recap & Checklist

Similarity Foundations

- **Similar Figures:** Same shape, not necessarily same size. All congruent figures are similar.
- **Thales Theorem:** Parallel lines divide triangle sides in the same ratio.

Similarity Criteria

- **AAA / AA:** Corresponding angles equal.
- **SSS:** Corresponding sides proportional.
- **SAS:** One angle equal and including sides proportional.

THANK YOU

"Geometry has two great treasures: one is the Theorem of Pythagoras, and the other the division of a line into extreme and mean ratio." — Johannes Kepler

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